

Smart Calorie Mixer

This project presents the development of a Calorie Burner system that uses an ESP8266 microcontroller and a Hall Effect sensor to monitor real-time cycling metrics such as speed and calories burned. The system is designed to offer a low-cost, offline, and educational alternative to commercial fitness trackers. By capturing pedal rotations using a magnetic sensor setup, the Calorie Burner provides live performance feedback on a 2.4" TFT LCD display, enabling users to track their cycling effort without the need for mobile apps or internet connectivity.

The core functionality includes calculating speed based on the time interval between sensor pulses and estimating calories burned using a dynamic MET (Metabolic Equivalent of Task) model, which adapts to the cyclist's speed. These real-time calculations provide accurate and personalized fitness data, enhancing the exercise experience. The system supports customizable inputs, such as user weight, and incorporates features like session summaries, max/average speed, and interrupt-based button controls for a user-friendly interface.

Calorie burner not only promotes a healthy lifestyle but also acts as a valuable educational platform for students and enthusiasts learning embedded systems. The project integrates concepts such as sensor interfacing, microcontroller programming, real-time data processing, and UI development. Its offline capability and open-source nature make it highly adaptable for academic use, DIY innovation, and future expansion into IoT-based health monitoring system.

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